

Take-Home Case: Write-up

Associate, People Intelligence · Fanatics Collectibles · Anushree · data as of September 1, 2026

What you're getting: (1) a live app, <https://fanatics-people-intelligence.streamlit.app/>, the interactive tool an HRBP would open (pick a leader, filter, click into sub-orgs, change the as-of month; upload next month's workbook to refresh it), with docs/People_Intelligence_Explorer.html in the repo as a backup that opens without any server; (2) the code on GitHub, https://github.com/anushreeayyar/Fanatics_People_Intelligence, with the cleaning code, metric engine, SQL cross-check, issue log and tests; (3) this note.

1. Answers (FTEs, as of Sept 1, 2026)

Q	Question	Answer
1	FTE headcount, Morgan Reyes' org	Morgan Reyes' organization has 418 FTEs as of September 1, 2026 (Morgan plus everyone below him). 417 if Morgan is excluded.
2	Annualised TCC, Dana Whitfield's org, FTEs	\$45,439,575 (552 FTEs incl. Dana): base \$44.05M + target bonus \$1.39M. \$45,122,875 (551 FTEs) excluding Dana. Excludes 9 people in the org with a 4% increase dated Oct 1 (about \$42K a year once effective).
3	Headcount trend this year; are we growing?	Headcount continued to grow through August, but growth slowed and August was the first month-over-month decline. Company FTEs 1,935 (Dec 31) to 1,991 (+56, +2.9%). Net adds: Jan +5, Feb +13, Mar +10, Apr +7, May +10, Jun +4, Jul +8, Aug -1. Hires flat at 22-30 a month while exits rose from 12 to 26.
4	Is voluntary attrition getting better or worse?	Worse. Annualised voluntary rate by quarter: Q4'24 5.5%, Q1'25 7.6%, Q2'25 7.7%, Q3'25 7.6%, Q4'25 7.1%, Q1'26 8.4%, Q2'26 10.1%, Q3'26 12.4% (July-August only). Trailing 12 months: 7.0% (Aug '25) to 9.2% (Aug '26). Q3'24 is omitted: the termination log starts mid-August 2024, so it has one month of data.
5	Are early-career employees leaving faster? Worse? Recommendation?	Yes, and it is worsening. Definition: voluntary FTE exits with under 365 days of service, divided by average FTE headcount under 365 days of service, annualised (74 exits over Sep 2025-Aug 2026 on an average of about 269 first-year FTEs). It is 27.5% over the last 12 months, versus 6.3% for everyone else. A different, cohort-style measure (exits within a year of hire divided by hires) gives lower figures and is shown as the hire-cohort table in the tool. A year ago the first-year rate was 13.0%. By quarter: about 13% through 2025, 24.8% in Q1'26, 38.3% in Q2'26, 40.5% in Q3'26 so far. See section 3.

Judgement calls to confirm: leader included in their own org; termination date treated as last day worked; first year = under 365 days (this ties to the team's monthly_summary tab); attrition = voluntary exits divided by average headcount, annualised.

2. Data issues, assumptions and how I handled them

Each issue is logged in data_issues_log.csv and shown in the tool's Data & method tab. The ones that change an answer:

- **Duplicate rows and a duplicate person.** Three exact duplicate rows were dropped. "Rob Calloway" (id 3994) and "Robert Calloway" (id 11973) share email, birthday, department and location, and Rob has no pay record, so Rob is excluded. That single record sits in Morgan's org and is exactly the 1-FTE gap between my recount and the monthly_summary every month.
- **Stale status.** Jordan Beck is marked Active but the termination log shows a July 15, 2026 resignation. The log wins.

- **Rehire under one ID.** ID 7532 has two stints (left Aug 2024, rejoined Nov 2025). Kept both spells; counted once.
- **Non-FTEs in the termination log.** All 76 "End of Internship" exits are coded Voluntary. Unfiltered, Aug 2025 shows 64 voluntary exits instead of 12. Attrition is FTE-only.
- **Pay.** Hourly rates for production staff annualised at 2,080 hours (assumption, no hours field). GBP/EUR/CAD converted at the single static rate in legal_entities. Latest record effective on or before the as-of date used; 40 rows dated Oct 1, 2026 excluded. Summing raw amounts would understate Dana's org by about \$11M (hourly) and over-count Morgan's by about \$42K (future-dated rows).
- **Formatting.** Two date formats plus padded dates (US M/D/YYYY confirmed because day values reach 31 in the second field); mixed-case status and department codes (lower-case codes fail a join); trailing spaces in locations.
- **Placeholder hire date.** 63 unique people (64 rows before removing a duplicate) have 2015-01-05. Left as is; none affect the under-1-year cohort, but tenure metrics for them are unreliable.
- **monthly_summary tab.** Headcount and under-1-year counts tie to my rebuild once the duplicate is removed; voluntary_terms differs in March 2026 (16 vs 13 in the log). I did not use it for any answer.

What a naive query would have returned for Morgan's org: 414 (status compared case-sensitively), 421 (any-case Active, duplicates and stale status kept), or 4 (direct reports only).

3. Question 5 in more detail, and question 6

- **Two problems, not one.** Outside Morgan Reyes' org, company voluntary attrition is roughly flat (7.4% vs 6.4% a year ago) but first-year attrition there rose from 12% to 21%. Inside Morgan's org, first-year attrition is 55% (30 exits) and tenured attrition is 10% vs 5%; all four director teams are above the company first-year rate. It is the only VP org that shrank this year (-22 vs +78 for the rest).
- **Why.** Among first-year leavers, relocation (30%) and return to school (20%) lead; compensation is 16%. Early leavers' base pay is close to peers at the same level (about 0.95 vs 0.96 of level median). The log cannot tell us more.
- **Recommendation.** (1) Review the last 20 first-year exits in Morgan's org case by case with its HRBP. (2) Run 30/90/180-day check-ins for everyone hired since January; the newest cohorts are leaving sooner (5% quit within 90 days for Q1'26 hires, 11% for Q2'26 vs about 3% earlier). (3) Do not default to a pay fix; look at who we hire (local candidates, expected-tenure screening) and career-path clarity. (4) Add exit-interview themes, hiring source and hiring manager to the data.

Q6: other things worth flagging

- Growth is stalling: August was the first month in the file where FTE headcount fell, and the last three months averaged 27 hires against 23 exits.
- Monthly voluntary exits went from 8-12 in the first half of the data to 20, 19 and 22 in Jun-Aug 2026.
- Forty people have a 4% base increase dated Oct 1 (about \$165K a year); only 40 of about 2,000 FTEs, so it looks like a partial load. Confirm with Total Rewards before Finance forecasts from this file.
- Record hygiene (duplicate person, stale status, ID reuse, placeholder hire date) each moves a headcount by one and should be fixed at source in HRIS.

4. AI appendix

Where I used AI. I worked with Claude (Anthropic) throughout: profiling the seven sheets, finding the data issues, writing the cleaning code, the JavaScript metric engine, the HTML tool and a first

draft of this note. I made the judgement calls (org definition, first-year definition, which duplicate to drop, treating the log as truth over status).

How the output was verified. (1) Three implementations of Q1 and Q2 agree to the cent: pandas, the JavaScript engine that powers the tool, and SQL (a recursive CTE, answers.sql). They share the cleaned tables, so this tests logic, not cleaning. (2) I rebuilt all 24 months of the monthly_summary from source and reconciled them. (3) I checked the tool in a headless browser on desktop and phone widths for errors. (4) I sanity-checked numbers against raw counts (for example, the hierarchy is a strict five-layer tree, every leader's org roll-up is complete, and no terminated manager has reports).

What had to be corrected along the way. The first recount was one FTE higher than the summary in every month; that led to finding the duplicate person. My first draft of the tenure cut used calendar years; changing to 365 days made the under-1-year counts tie exactly. Early drafts of the tool's narrative had hard-coded figures and one garbled sentence; I made them compute from the data. AI also does not know which of your judgement calls Finance or HR would want, so those are listed in section 1 for you to confirm.

5. If I had a week: fully automated, monthly, no one touching it

- **Source.** Replace the workbook with a scheduled extract from Oracle HCM into Snowflake (employees, hierarchy, comp, terminations) and load raw tables untouched.
- **Transform.** Port clean.py to SQL/dbt models: staging (formats, case, dedupe), a person-spell table that handles rehires and duplicate people, and an effective-dated comp model. dbt tests for the issues found here: one active row per person, status agrees with terminations, no future-dated comp in current views, currency matches entity, hierarchy has one root and no cycles.
- **Metrics.** Month-end headcount, attrition and TCC as governed tables with a documented data dictionary and FX from a dated rate table instead of a static rate.
- **Delivery.** A scheduled job (Airflow or GitHub Actions) runs on the 2nd business day, rebuilds the tool, and posts it to an internal site behind SSO.
- **Access and privacy.** Row-level security so each HRBP sees only their leaders' orgs; small-group masking beyond the 5-person rule already in the tool; no individual pay in exports; audit log of who viewed what.
- **Monitoring.** Automatic reconciliation against last month (headcount bridge: prior + hires - exits = current), alerts to the People Intelligence channel when a data test fails, and an AI-agent step that drafts the "what changed" summary for human review before release.
- **Retire the hand-kept summary tab** in favour of the pipeline output.

6. Time spent

Approximately 2 hours 15 minutes in total, split roughly as follows.

- **Data exploration, cleaning and validation:** about 25 minutes.
- **Metric definitions, calculations and reconciliation:** about 30 minutes.
- **Dashboard development and testing:** about 60 minutes.
- **Write-up, AI appendix and final QA:** about 20 minutes.